

Barriers for Learning in an Evolving World

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The Phenomenon: Loss of Plasticity (LoP)

The Continual Learning Problem

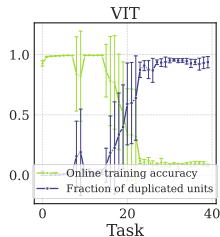
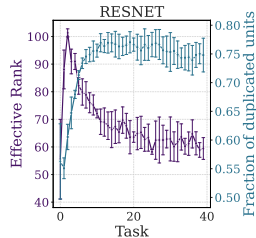
In continual streams, networks retain old knowledge but **lose the ability to fit future data**.

The Empirical Symptoms

Across architectures, as accuracy drops:

- **Dead units** rise (e.g., ResNet)
- **Duplicate features** emerge (e.g., ViT)
- **Effective rank collapses**

Why does GD fail to recover once this state appears?



The Hypothesis: LoP as a Topological Trap

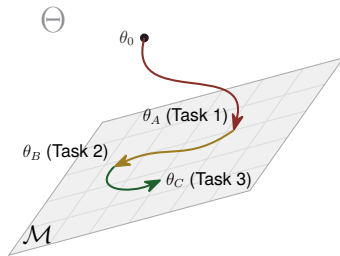
Prior work catalogs *symptoms*. We mathematically prove the **mechanism of persistence**.

A manifold $\mathcal{M} \subset \Theta$ induces LoP if the gradient is permanently tangent to it:

$$\nabla_{\theta} \ell(\theta; x, y) \in T_{\theta} \mathcal{M} \quad \forall (x, y)$$

Key Insight: The lock is enforced by the *network equations*, not the data. Once inside, discrete GD/SGD updates remain trapped in $\mathcal{M}$ forever, completely independent of the data distribution.

SEQUENTIAL TASKS (TRAPPED)



Theorem 1: The Exact Mechanics of Entrapment

Degeneracy in the computation $\Rightarrow$ degeneracy in the parameter update.

1. Frozen Manifold ($\mathcal{M}_F$)

If $f'(z_v) = 0$ persistently, incoming gradients vanish for every sample.

A frozen unit is permanently silent to backprop.

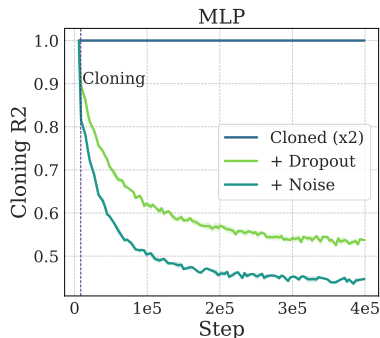
2. Cloning Manifold ($\mathcal{M}_C$)

If block pairs satisfy equal row/col sums, $h_u = h_{u'}$ and $\delta_v = \delta_{v'}$ strictly.

Cloned directions are indistinguishable.

Gradients locked forever.

Empirical Proof: SGD & Adam are Trapped



Even Adam is trapped. SGD and Adam achieve $R^2 = 1.0$ for 400k steps. Escape requires explicit symmetry breaking.

The Cause: The Rank-Plasticity Tension (Theorem 2)

Why do models fall in? Training balances **compression** (for generalization) and **nonlinear creation** (for diversity). *The very mechanisms that help a model generalize on a static task actively drive it into LoP manifolds.*

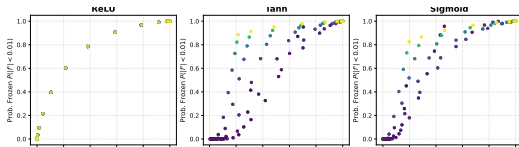
Rank-Gain Theorem

$$\text{Feature Creation} = \text{Strength } (\gamma) \times \text{Opportunity } (\Psi)$$

The Edge States of Feature Learning:

- Maximize strength $\gamma \uparrow$ causes $\mathcal{M}_F$ (**Frozen Trap**).
- Compress opportunity $\Psi \downarrow$ causes $\mathcal{M}_C$ (**Cloned Trap**).

Empirical Proof of the Tension

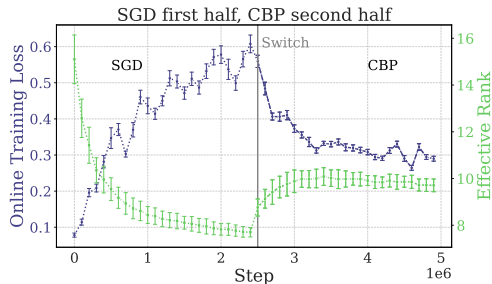


Regimes maximizing decorrelation strength drive units toward a high probability of freezing (proven across 6 activations).

Mitigation & Recovery

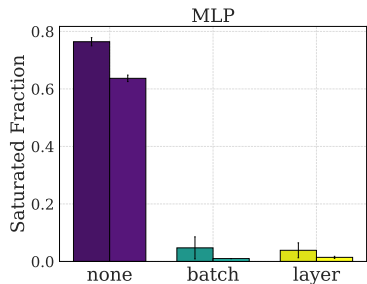
To restore plasticity, we must actively break symmetries or bound drift.

1. Escaping the Cloned Trap ($\mathcal{M}_C$)



Continual Backprop (CBP) injects targeted noise to break symmetry and successfully recover effective rank.

2. Preventing the Frozen Trap ($\mathcal{M}_F$)



Normalization layers bound pre-activation drift, preventing units from falling into the saturation zones.

Thank you!

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Paper: *Barriers for Learning in an Evolving World:
Mathematical Understanding of Loss of Plasticity*

Code: <https://github.com/ajoudaki/loss-of-plasticity>